

REDOX REACTIONS

MASTER STUDY GUIDE

Source-grounded • decision-first • subtopic-audited • cumulative transfer ready

6 SUBTOPICS

- 01 Oxidation Number
- 02 Oxidation & Reduction
- 03 Oxidising / Reducing Agents
- 04 Redox vs Non-redox
- 05 Types of Redox Reactions
- 06 Integrated Decision Pipeline

HARD BOUNDARY

No titration, normality/equivalent-weight machinery, full ion-electron balancing, electrochemistry or reaction-specific inorganic theory is added.

Chapter roadmap

Prerequisites first; ExamSIDE changes practice weight, not concept order.

ROADMAP

01 Oxidation Number

Assign correctly

02 Oxidation & Reduction

Track up/down or e^- loss/gain

03 Oxidising / Reducing Agents

Attach process to reactant species

04 Redox vs Non-redox

Prove ON change

05 Types

Classify topology after redox is proved

06 Integrated Decision Pipeline

Choose the minimum justified path

CANONICAL PIPELINE

SELECT -> ASSIGN -> TRACK -> CLASSIFY -> ATTACH -> NAME -> CHECK

Chapter source-obligation audit

Every chemistry obligation from the supplied PDF has a teaching home.

SOURCE

ID	Source obligation	Primary teaching home
RX-S01	Oxidation number meaning / bookkeeping charge	RX-ST01
RX-S02	Free element oxidation number = 0	RX-ST01
RX-S03	Group 1 / Group 2 fixed values	RX-ST01
RX-S04	Hydrogen default + hydride exception	RX-ST01
RX-S05	Oxygen default + peroxide/superoxide/O-F exceptions; F = -1	RX-ST01
RX-S06	Monatomic ion oxidation number = charge	RX-ST01
RX-S07	Sum of oxidation numbers = species charge	RX-ST01
RX-S08	Maximum-state check + metal-carbonyl source rule	RX-ST01
RX-S09	Average vs actual/non-equivalent oxidation states	RX-ST01
RX-S10	Oxidation = electron loss / ON increase	RX-ST02
RX-S11	Reduction = electron gain / ON decrease	RX-ST02
RX-S12	Oxidising agent definition	RX-ST03
RX-S13	Reducing agent definition	RX-ST03
RX-S14	Redox vs non-redox recognition	RX-ST04
RX-S15	Combination: redox and non-redox contrast	RX-ST05
RX-S16	Decomposition: redox and non-redox contrast	RX-ST05
RX-S17	Displacement redox pattern	RX-ST05
RX-S18	Disproportionation	RX-ST05
RX-S19	Comproportionation	RX-ST05
RX-S20	Intermolecular vs intramolecular redox	RX-ST05

SOURCE CONTENT STATUS

20/20 obligations taught. One source-internal practice item ('strongest reducing metal') is separately flagged as needing reactivity/periodicity knowledge that the source theory does not teach.

REDOX REACTIONS

SUBTOPIC 01 - OXIDATION NUMBER

Meaning -> rules -> exceptions -> sum rule -> average vs actual -> transfer

THE QUESTION

Given a formula or ion, how do I assign the oxidation number without guessing?

BOOK RHYTHM

Familiar context -> explanation -> visual helper -> worked example -> misconception repair -> guided practice -> check.

Source boundary: oxidation-number content from the supplied Redox reaction.pdf only.

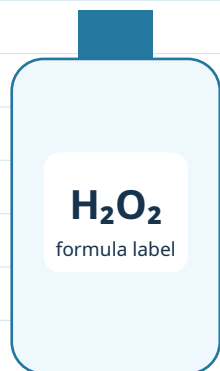
Context examples are pedagogical bridges; they do not add a new chemistry unit.

What is an oxidation number?

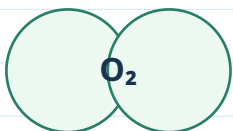
Understand the bookkeeping idea before memorising rules.

ON-C1

A familiar formula can hide a rule



You may recognise the formula H_2O_2 on a hydrogen-peroxide label. The formula looks simple, but oxygen is not -2 here; in a peroxide it is -1. That is why oxidation numbers are not found by guessing from the element alone.



By contrast, oxygen in elemental O_2 has oxidation number 0. The same element can therefore have different oxidation numbers in different species.

WHAT THE NUMBER MEANS

An oxidation number is a bookkeeping number: the hypothetical charge an atom would have if bonding were treated ionically. It may be positive, negative or zero.

The purpose of the number

Later in Redox, you will compare an atom's oxidation number before and after a reaction. For now, the job is simpler: learn to assign the

FOR EXAMPLE

HCl : $\text{H} = +1$ and $\text{Cl} = -1$. The molecule is neutral, so the assigned values add to zero.

CCl_4 : four Cl atoms contribute -4 in total, so carbon must be +4.

CONCEPT HELPER

Think of oxidation numbers as a charge-accounting system. Your assigned values must reproduce the total charge of the species.

MISCONCEPTION CLINIC

Wrong model: oxidation number is always the atom's real ionic charge.
Repair: it is a bookkeeping assignment and may be an average value.

Read the formula before you calculate

Subscripts count atoms; superscripts show charge.

ON-C2

Formula anatomy



2, 7 = atom counts

²⁻ = ion charge

In $\text{Cr}_2\text{O}_7^{2-}$, the small lower numbers tell you how many atoms are present. The raised 2- is the charge of the whole ion. These are different pieces of information.

MISCONCEPTION CLINIC

Wrong model: treat the subscript 2 or 7 as an oxidation number.
Repair: subscripts count atoms; oxidation numbers are assigned separately.

QUICK CHECK

In NO_3^- , what does 3 mean? What does - mean?

Rule-priority ladder

- 1 Free element? ON = 0
- 2 Monatomic ion? ON = charge
- 3 Fixed rule? Group 1 / 2, F
- 4 H or O exception? check first

WHY A PRIORITY LADDER HELPS

It prevents two common errors: doing algebra when a fixed rule already gives the answer, and using the normal O/H rule before checking an exception.

SOURCE-SPECIFIC RULE

For the metal-carbonyl examples in the source, the metal oxidation number is taken as 0: e.g. Ni in $\text{Ni}(\text{CO})_4$.

The small fixed set

Use these before the sum rule.

ON-C3

Rules worth memorising

Free element

O_2 , Cl_2 , S_8 , Cu

0

Group 1 metals

Li, Na, K ...

+1

Group 2 metals

Mg, Ca ...

+2

Fluorine

in compounds

-1

FAST RECOGNITION

If K appears in K_2O , K_2O_2 or KO_2 , potassium is +1 in all three because the Group 1 rule is fixed.

Check Your Knowledge

- 1 Oxidation number of O in O_2 ?
- 2 Oxidation number of Na in Na^+ ?
- 3 Oxidation number of Mg in MgO ?
- 4 Oxidation number of K in KO_2 ?
- 5 Oxidation number of Fe in $Fe(CO)_5$ using the source rule?

ANSWERS

0; +1; +2; +1; 0.

Hydrogen and oxygen exceptions

Ordinary rule first; exception second.

ON-C4

Hydrogen

Hydrogen is normally +1. In metal/ionic hydrides, H = -1.

COMPARE

CH₄: H = +1
LiH / NaH: H = -1

FIRST QUESTION

Before assigning H, ask: ordinary covalent compound or metal hydride?

MISCONCEPTION CLINIC

Wrong model: H is always +1.
Repair: metal hydrides are the explicit source exception.

Oxygen

ordinary O -2
H₂O, MgO

peroxide O -1
H₂O₂, Na₂O₂

superoxide O -1/2
KO₂

O bonded to F positive
O₂F₂: +1; OF₂: +2

MISCONCEPTION CLINIC

Wrong model: oxygen is always -2.
Repair: identify peroxide, superoxide, or O-F cases before using the default.

The master sum rule

Assigned values must reproduce the species charge.

ON-C5

Neutral molecule

Concept helper: make every atom add to the species charge

2H +2

S x

4O -8

TOTAL 0

WORKED: H_2SO_4

Known: H = +1, O = -2.

$$2(+1) + x + 4(-2) = 0$$

$$2 + x - 8 = 0 \rightarrow x = +6.$$

CHECK

+2 +6 -8 = 0. The assigned values reproduce a neutral molecule.

Polyatomic ion

Concept helper: make every atom add to the species charge

2Cr 2x

7O -14

TOTAL -2

WORKED: $\text{Cr}_2\text{O}_7^{2-}$

Known: O = -2.

$$2x + 7(-2) = -2$$

$$2x - 14 = -2 \rightarrow x = +6.$$

MISCONCEPTION CLINIC

Wrong model: every sum equals zero.

Repair: an ion's sum equals its written charge.

One method, three different formulas

Read -> place fixed values -> sum -> solve -> check.

ON-C6

Worked examples



$$2(+1) + 2x + 7(-2) = 0$$
$$2 + 2x - 14 = 0$$
$$\text{Cr} = +6$$



$$(+1) + x + 4(-2) = 0$$
$$1 + x - 8 = 0$$
$$\text{Mn} = +7$$



$$x + (-2) = +2$$
$$\text{V} = +4$$

Ion charge changes the right-hand side.

Concept helper: the 5-move routine

- 1 Read formula / charge
- 2 Place fixed values
- 3 Check H/O exceptions
- 4 Set sum = 0 or charge
- 5 Solve + substitute back

TRY WITH ME

H_2SO_3 : $2(+1) + x + 3(-2) = 0$.
Solve x, then substitute it back to verify the total.

Answer: S = +4.

The same element can have many oxidation states

Calculate first; order or compare second.

ON-C7

Nitrogen: convert formulas into numbers



+1 avg



+2



+3 avg



+4



+5

CONCEPT HELPER

For ordering questions, do not compare formulas directly. Convert each formula to a number first; then sort the numbers.

Phosphorus: one formula family



$$4(+1) + 2x + 7(-2) = 0$$
$$\text{P} = +5$$



$$4(+1) + 2x + 5(-2) = 0$$
$$\text{P} = +3$$



$$4(+1) + 2x + 6(-2) = 0$$
$$\text{P} = +4$$

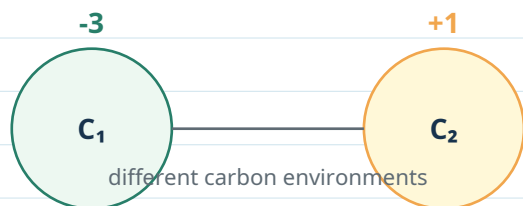
Average oxidation number vs actual atoms

A fractional average does not mean every atom is fractional.

ON-C8

The problem that creates the idea

In CH_3CHO , the two carbon atoms are not equivalent. The source assigns them different oxidation numbers.



AVERAGE CARBON

Average = $[-3 + (+1)] / 2 = -1$. The average is useful, but it is not the actual value of either carbon site.

MISCONCEPTION

Fractional/average value every atom has that value.

Mixed-valence example: Fe_3O_4

AVERAGE FIRST

$$3x + 4(-2) = 0 \rightarrow x = +8/3$$

BUT THE SOURCE SHOWS

$\text{FeO} + \text{Fe}_2\text{O}_3$: Fe(II) and Fe(III) components are both present. So $+8/3$ is an average across different Fe sites.

RETRY CHECK

If a question asks for average Fe, $+8/3$ is fine. If it asks for the actual site states, answer +2 and +3.

ON-C9

Two source examples



The source representation has chlorine at -1 and +1. The formula-average Cl value is 0, but the two actual sites are not both 0.



The source shows terminal sulfur atoms at +5 and inner sulfur atoms at 0. Average sulfur = +2.5.

FIRST QUESTION

Does the problem ask for an average, or for individual non-equivalent atoms?

The CrO_5 trap

The source structure contains peroxide oxygen atoms. A blind $\text{O} = -2$ rule would fail.

CONCEPT HELPER

Four O atoms are peroxide O: each -1. One terminal oxo O is -2.

NOW CALCULATE

$$\begin{aligned}x + 4(-1) + 1(-2) &= 0 \\x &= +6\end{aligned}$$

MISCONCEPTION REPAIR

Structure can override a default rule by telling you which oxygen atoms are peroxide oxygens.

Practice: full scaffold -> faded scaffold

Build independence before ExamSIDE transfer.

ON-C10

Try with me

A. $\text{Cr}_2\text{O}_7^{2-}$

1. O = -2.
2. Ion total = -2.
3. Write $2x + 7(-2) = -2$.
4. Solve and check.

B. OF_2

1. F = -1.
2. Neutral total = 0.
3. Write $x + 2(-1) = 0$.
4. Solve.

ANSWERS

A: Cr = +6. B: O = +2.

Now you try - no setup given

- 1 Find Mn in KMnO_4 .
- 2 Find Fe in K_2FeO_4 .
- 3 Find O in H_2O_2 .
- 4 Find O in KO_2 .
- 5 Find average Fe in Fe_3O_4 .
- 6 Find N in NO_3^- .

ANSWER STRIP

+7; +6; -1; -1/2; +8/3 average; +5.

AUDIT

Learner self-check

- 1 I can distinguish a subscript from an ionic charge.
- 2 I use the rule-priority ladder before algebra.
- 3 I check H/O exceptions before the default rule.
- 4 I set the oxidation-number sum equal to 0 or to the ion charge correctly.
- 5 I can calculate an unknown oxidation number and verify it.
- 6 I can distinguish an average oxidation number from actual non-equivalent atoms.

SOURCE QC NOTE

The supplied scan's CH₄ carbon table entry is internally inconsistent. Using the source's own rules gives C = -4; this correction stays inside the same scope.

Subtopic completeness audit

SOURCE_OBLIGATIONS

PASS

FAMILIAR / REAL-LIFE BRIDGE

PASS

CONCEPT HELPERS

PASS

MISCONCEPTION REPAIRS

PASS

WORKED + GUIDED + FADED PRACTICE

PASS

FORMULA TYPOGRAPHY

PASS

NEXT

ExamSIDE transfer for this subtopic is audited separately against the canonical required question set.

REDOX REACTIONS

SUBTOPIC 02 - OXIDATION & REDUCTION

Electron loss/gain and oxidation-number up/down are the same story

THE QUESTION

When a species changes, can you decide oxidation or reduction from either electrons or oxidation number?

BOUNDARY

This guide stops before oxidising agent / reducing agent naming. Those roles belong to Subtopic 03.

Source-grounded to the supplied Redox reaction.pdf oxidation/reduction definitions and practice.

One change, two languages

Electron loss/gain and oxidation-number change must agree.

OR-C1

Oxidation

SOURCE EXAMPLE: Mg



Mg loses 2 electrons. Its oxidation number changes 0 $\rightarrow$ +2.
Both descriptions mean oxidation.

0 $\xrightarrow{\text{oxidation number UP}}$ +2

MEMORY LINE

LOSE e^{-} $\rightarrow$ oxidation $\rightarrow$ oxidation number UP

SOURCE WORD

Loss of electrons is also called de-electronation in the source practice.

Reduction

SOURCE EXAMPLE: Fe^{3+}



Fe^{3+} gains 1 electron. Its oxidation number changes +3 $\rightarrow$ +2.
Both descriptions mean reduction.

+3 $\xrightarrow{\text{oxidation number DOWN}}$ +2

MEMORY LINE

GAIN e^{-} $\rightarrow$ reduction $\rightarrow$ oxidation number DOWN

TYPOGRAPHY CHECK

Electron is written e^{-} ; ionic charges are raised: Fe^{3+} , Fe^{2+} .

Why losing a negative electron makes the number rise

The sign feels backward until you picture the charge.

OR-C2

Remove negative charge

Concept helper: electrons are negative



An electron carries negative charge. When a species loses electrons, it becomes less negative / more positive. That is why its oxidation number

MICRO-CONTRAST

$\text{Cl}^- \rightarrow \text{Cl} + \text{e}^-$: -1 $\rightarrow$ 0, so oxidation.

$\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$: 0 $\rightarrow$ +1, so oxidation.

MISCONCEPTION REPAIR

Wrong: losing a negative electron should make oxidation number fall.
Repair: removing negative charge makes the state more positive.

Add negative charge

Concept helper: electrons are negative



When a species gains electrons, it gains negative charge. Its oxidation number therefore becomes less positive / more negative: it decreases.

MICRO-CONTRAST

$\text{F} + \text{e}^- \rightarrow \text{F}^-$: 0 $\rightarrow$ -1, so reduction.

$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$: +2 $\rightarrow$ 0, so reduction.

MISCONCEPTION REPAIR

Wrong: gain means the oxidation number rises.
Repair: gaining negative charge makes oxidation number fall.

The same decision from electron notation or oxidation number

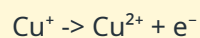
Use whichever cue is clearer.

OR-C3

Electron notation

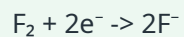
OXIDATION

Electrons are lost.



REDUCTION

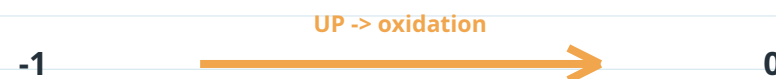
Electrons are gained.



CONCEPT HELPER

If the equation already shows e^- , decide loss or gain before thinking about any other label.

Oxidation-number notation



SECOND LANGUAGE

UP and DOWN are a cross-check for the electron story, not a separate topic.

FAST RETRIEVAL

UP = oxidation. DOWN = reduction.

How many electrons are involved?

Use the oxidation-number jump per atom, then count atoms.

OR-C4

Per atom first



I: $-1 \rightarrow 0$. The rise is 1, so each I^- loses 1e^- .



S: $-2 \rightarrow 0$. The rise is 2, so each S^{2-} loses 2e^- .

FIRST MOVE

Find the oxidation-number jump per atom. Then multiply only if the final species contains more than one atom of that element.

Then use atom count



Each I loses 1e^- , but I_2 contains two iodine atoms. Total electron loss = 2e^- .



Average C changes $+3 \rightarrow +4$. Each carbon loses 1e^- . One CO_2 contains one C, so 1e^- per CO_2 molecule.

SCOPE NOTE

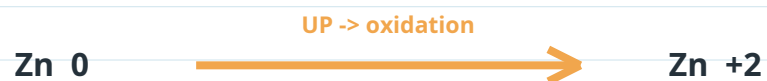
This is electron accounting from oxidation-number change, not a full redox-balancing lesson.

Oxidation and reduction happen together in a redox reaction

Two lanes describe one electron transfer.

OR-C5

Source-style equation



One species loses electrons while another gains them. In a complete redox reaction, oxidation and reduction occur simultaneously.

SELF-CHECK

If you have identified oxidation in a complete redox equation, look for a corresponding reduction before you stop.

Do not jump to agent names yet

THIS SUBTOPIC ASKS

What is oxidised? What is reduced? How many electrons are lost or gained?

NEXT SUBTOPIC ASKS

Which reactant is the oxidising agent and which is the reducing agent?

WHY SEPARATE THEM

Separating the process from the agent name prevents the common label inversion.

OR-C6

'Oxidation means adding oxygen.'

Electron loss / ON increase is the source definition used here. Oxygen need not appear.

'If an electron is on the right, it is reduction.'

Electron on the product side means the species lost it: oxidation.

'Reduction means removing oxygen.'

Electron gain / ON decrease is the source definition used here.

'A bigger positive ion means more reduction.'

Compare before $\rightarrow$ after. $+2 \rightarrow +3$ is oxidation; $+3 \rightarrow +2$ is reduction.

RETRY A

$\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$. Electron lost; $+2 \rightarrow +3$. Name the process.

RETRY B

$\text{Cl} + \text{e}^- \rightarrow \text{Cl}^-$. Electron gained; $0 \rightarrow -1$. Name the process.

OR-C7

A. Which is NOT oxidation?

Compare four changes

1. $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$
2. $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$
3. $\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^-$
4. $\text{Cl} + \text{e}^- \rightarrow \text{Cl}^-$

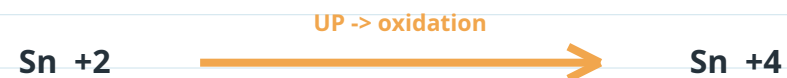
WORK THE CUE

1-3 lose electrons $\rightarrow$ oxidation. 4 gains an electron $\rightarrow$ reduction.

SOURCE CONNECTION

This mirrors the supplied PDF's CUQ distinction between oxidation and reduction equations.

B. Identify oxidation and reduction



ANSWER IN PROCESS LANGUAGE

Sn^{2+} is oxidised.
 Hg^{2+} is reduced.

DO NOT NAME AGENTS YET

That inversion belongs to Subtopic 03.

Now you try

Faded scaffold -> independent check.

OR-C8

Faded practice

- 1 $\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$: oxidation or reduction?
- 2 $\text{F} + \text{e}^- \rightarrow \text{F}^-$: oxidation or reduction?
- 3 $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$: oxidation or reduction?
- 4 $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$: oxidation or reduction?
- 5 N in $\text{NH}_4^+ \rightarrow \text{N}_2$: is nitrogen oxidised or reduced?

ANSWER STRIP

oxidation; reduction; reduction; oxidation; nitrogen is oxidised (-3 -> 0).

Independent reasoning

- 1 Why does losing e^- make oxidation number increase?
- 2 How many electrons are lost when 2I^- form I_2 ?
- 3 How many electrons are lost by one S^{2-} becoming S?
- 4 Carbon in oxalate changes +3 -> +4. How many e^- are lost per carbon?
- 5 Why should oxidation and reduction be found together in a complete redox reaction?

SOURCE VOCABULARY

Loss of electrons = oxidation = de-electronation in the source terminology.

Subtopic self-check

Source, pedagogy, typography and layout must all pass.

AUDIT

Learner checklist

- 1 I define oxidation as electron loss.
- 2 I define reduction as electron gain.
- 3 I connect oxidation to oxidation-number increase.
- 4 I connect reduction to oxidation-number decrease.
- 5 I can calculate e^- lost/gained from the oxidation-number jump.
- 6 I can include atom count when finding total electron change.
- 7 I can find both oxidation and reduction in one complete redox equation.

SOURCE PRACTICE COVERED

Definitions, equation recognition, de-electronation, Sn/Hg tracking, and nitrogen-oxidation cues are supported.

Completeness audit

SOURCE

PASS

PEDAGOGY

PASS

TRANSFER

PASS

CONTEXT BRIDGE

PASS

CONCEPT HELPERS

PASS

MISCONCEPTIONS

PASS

ASCII LEAKS

0

SUPERSCRIT FAILS

0

HEADER FAILS

0

LAYOUT FAILS

0

NEXT

Subtopic 03 attaches the oxidising-agent / reducing-agent names to these processes.

REDOX REACTIONS

SUBTOPIC 03 - OXIDISING & REDUCING AGENTS

The inversion that becomes easy once SELF and OTHER are separated

THE CORE QUESTION

Given an equation, which reactant is the oxidising agent and which is the reducing agent?

MAIN MISCONCEPTION

Students often reverse the labels because the agent is named by what it causes in the OTHER substance, not by what happens to itself.

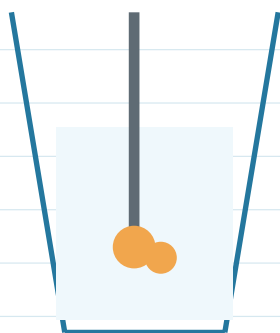
See the reaction before the labels

A familiar displacement reaction gives the whole story.

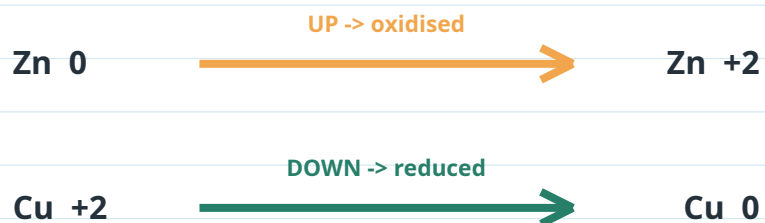
AR-C1

A visible change

In the source example, zinc placed in copper sulfate undergoes a redox change. The equation is $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$. The important question is not the blue colour or deposit itself; it is which reactant changes oxidation number in which direction.



Zn in CuSO_4



Now attach the agent names

REDUCING AGENT

- SELF** Zn is oxidised. It loses electrons.
- OTHER** Those electrons allow Cu^{2+} to be reduced.

OXIDISING AGENT

- SELF** Cu^{2+} is reduced. It gains electrons.
- OTHER** It causes Zn to be oxidised.

THE INVERSION

Oxidised species = reducing agent.
Reduced species = oxidising agent.

Why the names feel backwards

Separate SELF from OTHER.

AR-C2

Why the agent name is an inversion

An oxidising agent causes another species to undergo oxidation. To do that, it accepts electrons from the other species, so itself gets reduced.

A reducing agent causes another species to undergo reduction. To do that, it supplies electrons, so itself gets oxidised.

MEMORY PAIR

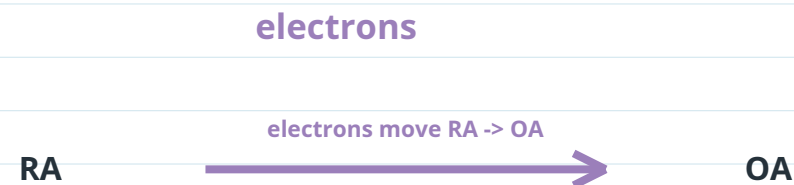
OA: SELF reduced -> OTHER oxidised

RA: SELF oxidised -> OTHER reduced

MISCONCEPTION CLINIC

Wrong model: oxidising agent must be oxidised.
Repair: agent name describes what it causes in the other reactant.

Two-arrow concept helper



FOLLOW THE ELECTRONS

The reducing agent gives electrons and is oxidised. The oxidising agent receives electrons and is reduced.

FAST RETRIEVAL

If oxidation number goes UP: that reactant is the reducing agent.
If oxidation number goes DOWN: that reactant is the oxidising agent.

The 6-move decision strip

Do the chemistry first; name the agents last.

AR-C3

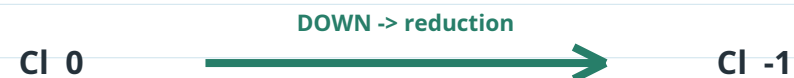
Use this on every equation

- 1 SELECT** changing atoms
- 2 ASSIGN** oxidation numbers
- 3 TRACK** before -> after
- 4 CLASSIFY** up = oxidation; down = reduction
- 5 ATTACH** to reactant species
- 6 NAME** oxidised species = RA; reduced species = OA

WHY THIS ORDER WORKS

Most wrong OA/RA answers happen when the learner tries to name the agent before proving what happened to the reactant itself.

Worked through the strip



ATTACH + NAME

Ca is the reactant carrying the oxidation -> reducing agent.
Cl₂ is the reactant carrying the reduction -> oxidising agent.

CHECK

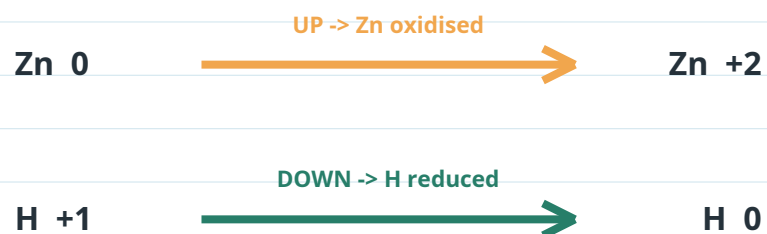
One oxidation and one reduction are present; the electron story is consistent.

Name the reactant species, not only the atom

The chemistry may be right but the MCQ label can still be wrong.

AR-C4

Example: $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$



WHAT SHOULD YOU NAME?

Zn is the reducing agent.
 H^+ is the oxidising species; in an MCQ whose options are whole reagents, HCl may be the expected reactant label.

SPECTATOR CHECK

Cl^- does not change oxidation number. Do not name a spectator ion as the redox agent.

A naming rule that prevents option mistakes

STEP 1 - IDENTIFY THE CHANGING ATOM

Use oxidation number to decide whether that atom is oxidised or reduced.

STEP 2 - ATTACH IT TO THE REACTANT

Name the reactant species that contains that changing atom, at the level used by the equation/options.

MISCONCEPTION CLINIC

Wrong model: 'H is the oxidising agent.'
Repair: H is the tracked atom; H^+/HCl is the reactant species.

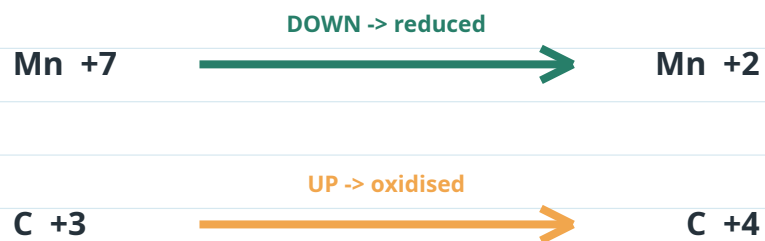
A multi-atom reagent still uses the same rule

Track only the atoms that actually change.

AR-C5

Permanganate + oxalic acid

The source practice includes the reaction of KMnO_4 with $\text{H}_2\text{C}_2\text{O}_4$ in acidic medium. You do not need a memorised oxidant/reductant list when the products are given: calculate the changing oxidation numbers.



NAME THE REAGENTS

KMnO_4 contains Mn that is reduced $\rightarrow$ oxidising agent.
 $\text{H}_2\text{C}_2\text{O}_4$ contains C that is oxidised $\rightarrow$ reducing agent.

Concept helper: ignore unchanged atoms first

SELECT

Circle only atoms whose oxidation state differs between reactant and product. K, H, sulfate-type spectators often do not need to be tracked for the agent decision.

TRACK

Write two tiny lanes: $\text{Mn} +7 \rightarrow +2$ and $\text{C} +3 \rightarrow +4$. Those two lanes are enough to assign both agents.

MISCONCEPTION CLINIC

Wrong model: calculate oxidation number for every atom in the equation.
Repair: select the likely changing atoms first.

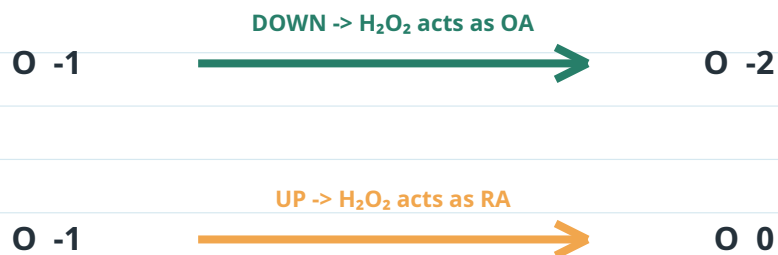
Can a species act as both types of agent?

Intermediate oxidation states can move in either direction.

AR-C6

Hydrogen peroxide is the cleanest source example

In H_2O_2 , oxygen has oxidation number -1. It can move down to -2 (reduction) or up to 0 (oxidation). So depending on the reaction, H_2O_2 can behave as an oxidising agent or a reducing agent.



CONCEPT HELPER

A species at an intermediate oxidation state may have room to be oxidised or reduced. The equation tells you which role it plays.

Extremes help with 'cannot act' questions

LOWEST STATE

If the redox-active element is already at its lowest available oxidation state in the source context, it cannot be reduced further, so it cannot function as an oxidising agent by that element.

HIGHEST STATE

If the redox-active element is at its highest source-supported oxidation state and the other atoms are fixed, the species cannot be oxidised further by that element, so it cannot act as a reducing agent.

EXAM USE

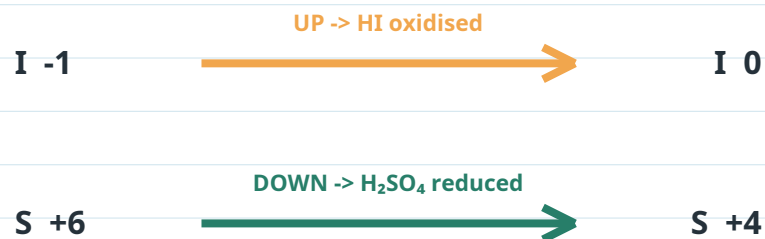
This supports questions such as N^{3-} 'cannot be an oxidising agent' and H_3PO_4 'cannot act both ways' without memorising reagent lists.

One reagent, one equation, one proof

Use oxidation-state evidence instead of reputation.

AR-C7

Oxidising behaviour of H_2SO_4



CONCLUSION

HI is the reducing agent because iodide is oxidised.
 H_2SO_4 is the oxidising agent because sulfur is reduced from +6 to +4.

NO MEMORISED LIST NEEDED

The equation itself proves the roles.

Close contrast



NO OXIDATION-STATE CHANGE

No tracked element changes oxidation number. This equation does not demonstrate oxidising behaviour of H_2SO_4 .

MISCONCEPTION CLINIC

Wrong model: a familiar oxidising reagent is always acting as an oxidising agent.
Repair: role belongs to the specific reaction.

FIRST MOVE

Prove an oxidation-number change before naming a role.

AR-C8

1. 'Oxidising agent is oxidised'

Reverse it: OA is reduced; it oxidises the other species.

2. 'Name the changing atom only'

Attach the changed atom to its reactant species.

3. 'Every ion in the reagent is an agent'

Spectator ions with unchanged oxidation state are not the redox agent.

4. 'Memorise KMnO_4 , H_2O_2 , H_2SO_4 roles'

A reagent's role can depend on the specific equation. Track the change.

RETRY A

$\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$. Which reactant is reduced? Therefore which is the oxidising agent?

RETRY B

$\text{H}_2\text{O}_2 \rightarrow \text{O}_2$. Oxygen goes $-1 \rightarrow 0$. Is H_2O_2 acting as OA or RA?

Practice: scaffold fading

Repeat the same pipeline until the inversion becomes automatic.

AR-C9

Try with me

A. $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$

Zn: 0 $\rightarrow$ +2 = oxidation $\rightarrow$ Zn is RA.
H: +1 $\rightarrow$ 0 = reduction $\rightarrow$ H⁺/HCl is OA.

B. $\text{Ca} + \text{Cl}_2 \rightarrow \text{CaCl}_2$

Ca: 0 $\rightarrow$ +2 = oxidation $\rightarrow$ Ca is RA.
Cl: 0 $\rightarrow$ -1 = reduction $\rightarrow$ Cl₂ is OA.

THE PATTERN

UP $\rightarrow$ oxidised $\rightarrow$ RA. DOWN $\rightarrow$ reduced $\rightarrow$ OA.

Now you try

- 1 $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$: identify OA and RA.
- 2 $2\text{HI} + \text{H}_2\text{SO}_4 \rightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$: identify OA and RA.
- 3 In $\text{H}_2\text{O}_2 \rightarrow \text{O}_2$, what role does H_2O_2 play?
- 4 Why can N^{3-} not function as an oxidising agent in the source-context reasoning?
- 5 Why is Cl^- a spectator in $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$?

ANSWER CUES

$\text{Cu}^{2+}/\text{CuSO}_4$ OA, Zn RA; H_2SO_4 OA, HI RA; H_2O_2 RA; N already at -3; Cl unchanged.

Subtopic self-check

Ready for redox recognition when the agent decision is mechanical.

AUDIT

Learner checklist

- 1 I know OA is itself reduced and RA is itself oxidised.
- 2 I can explain SELF vs OTHER rather than only recite a mnemonic.
- 3 I run SELECT -> ASSIGN -> TRACK -> CLASSIFY before naming agents.
- 4 I attach the role to the reactant species, not only the atom.
- 5 I ignore spectator species with unchanged oxidation number.
- 6 I can decide when an intermediate oxidation state can act in either direction.

SOURCE PRACTICE COVERED

The guide explicitly supports the supplied PDF's Zn/HCl, Ca/Cl₂, Zn/CuSO₄ and KMnO₄/oxalic-acid style agent-identification questions.

Completeness audit

SOURCE AGENT DEFINITIONS

PASS

FAMILIAR CONTEXT BRIDGE

PASS

SELF vs OTHER HELPER

PASS

DECISION STRIP

PASS

SPECIES-ATTACHMENT HELPER

PASS

MISCONCEPTION REPAIRS

PASS

NEXT

Next subtopic: prove whether an entire reaction is redox or non-redox from oxidation-state changes.

REDOX REACTIONS

SUBTOPIC 04 - REDOX vs NON-REDOX

Do not classify from appearance - prove an oxidation-number change

THE QUESTION

Given a complete chemical equation, can you prove whether it is a redox reaction or a non-redox reaction?

THE CORE TEST

A reaction is redox when oxidation numbers change: one tracked species is oxidised and another is reduced. If no oxidation number changes, it is non-redox.

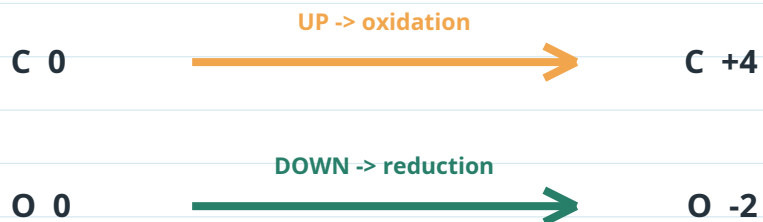
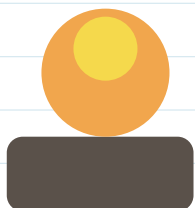
Two reactions can look very different - the oxidation numbers decide

Use a familiar contrast before the formal test.

RR-C1

Burning carbon

The source uses $C + O_2 \rightarrow CO_2$. Carbon and oxygen both change oxidation number, so this is a redox reaction.

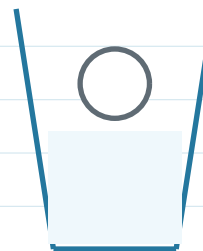


VERDICT

REDOX: there is an UP and a DOWN.

Quicklime + water

The source contrasts this with $CaO + H_2O \rightarrow Ca(OH)_2$. Ca stays +2, H stays +1 and O stays -2. No oxidation number changes.



VERDICT

NON-REDOX: atoms are rearranged into a new compound, but no oxidation number changes.

MISCONCEPTION CLINIC

Wrong model: every chemical change is redox.
Repair: redox requires oxidation-number change, not merely new products.

The redox fingerprint

Prove the change instead of guessing from the reaction label.

RR-C2

A five-move recognition helper

- 1 Choose atoms likely to change.
- 2 Write oxidation number before $\rightarrow$ after.
- 3 Any oxidation-number change?
- 4 If NO $\rightarrow$ non-redox.

5 If YES $\rightarrow$ find one UP and one DOWN $\rightarrow$ redox

THE FINGERPRINT

REDOX: at least one oxidation number rises and another falls in the complete reaction.
NON-REDOX: all tracked oxidation numbers remain unchanged.

DO NOT OVER-CALCULATE

Track only atoms likely to change; unchanged spectator atoms need not all be computed.

Why UP and DOWN should appear together

ELECTRON CONSERVATION STORY

If one species is oxidised, it loses electrons. In the complete redox reaction, another species must gain those electrons and be reduced.

SELF-CHECK

If you found only one oxidation-number change, either the equation is incomplete or you have probably missed the second changing element.

FAST TEST

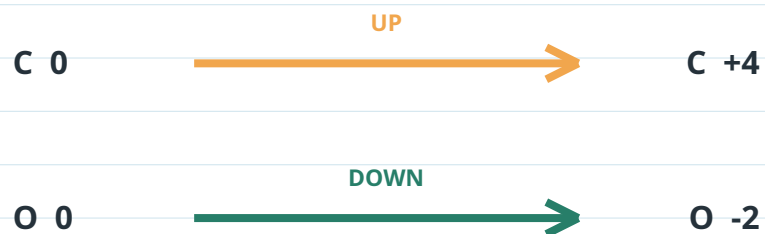
No ON change $\rightarrow$ non-redox. Never call a reaction redox solely because oxygen, heat, colour or gas appears.

Combination does not automatically mean redox

The source gives one redox and one non-redox combination example.

RR-C3

Combination + REDOX



PROOF

Both C and O change oxidation number. Therefore the reaction is redox.

IMPORTANT

Reaction type = combination. Redox status = redox. These are separate labels.

Combination + NON-REDOX



TRACK

Ca: +2 $\rightarrow$ +2
H: +1 $\rightarrow$ +1
O: -2 $\rightarrow$ -2

PROOF

No oxidation number changes. Therefore this combination reaction is non-redox.

MISCONCEPTION

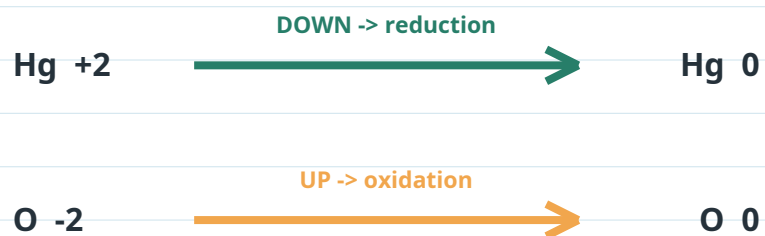
'Combination = redox' is false.

Decomposition also needs an oxidation-number check

A reaction splitting apart may or may not be redox.

RR-C4

Decomposition + REDOX



PROOF

Mercury is reduced and oxygen is oxidised. Therefore the decomposition is redox.

NOTE

Both changing atoms came from the same compound; classification by location comes later.

Decomposition + NON-REDOX



TRACK

Mg remains +2.
C remains +4.
O remains -2.

PROOF

No oxidation number changes. Therefore this decomposition reaction is non-redox.

MISCONCEPTION

'Decomposition = redox' is false.

A new form of the same element can still be non-redox

Elemental forms keep oxidation number 0.

RR-C5

Transfer trap: $O_2 \rightarrow O_3$



OXIDATION NUMBER

O in elemental O_2 = 0. O in elemental O_3 = 0.

VERDICT

0 $\rightarrow$ 0. No oxidation-number change, so this is non-redox even though one allotrope changes into another.

MISCONCEPTION CLINIC

Wrong: new molecular formula means redox.
Repair: elemental forms have oxidation number 0.

Transfer trap: neutralisation / precipitation



Acid-base neutralisation can form new products with no oxidation-number change.



A precipitate may form, but Ag remains +1 and Cl remains -1. Visible change alone does not prove redox.

CONCEPT HELPER

Ask what changed electronically?, not merely what changed visibly?

How to scan four equations quickly

Use likely-changing atoms before full oxidation-number algebra.

RR-C6

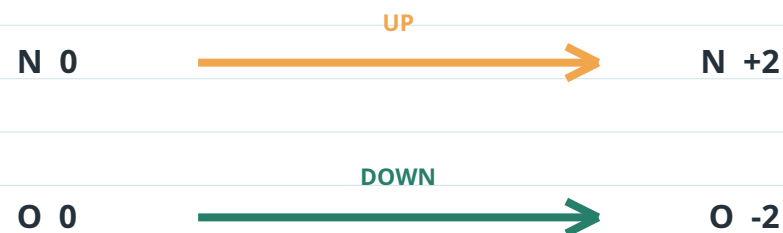
Fast MCQ workflow

- 1 Spot free elements: metals, O_2 , N_2 , halogens often start at 0.
- 2 Spot ions/elements in products: elemental metals/halogens are 0.
- 3 Track only the element likely to change across both sides.
- 4 If all likely atoms keep their states, reject as non-redox.
- 5 When one option shows both an UP and a DOWN, verify and select.

WHY THIS IS FASTER

Most JEE recognition options are built from obvious non-redox exchanges plus one equation containing elemental/reactive species that genuinely change oxidation number.

Worked scan: $N_2 + O_2 \rightarrow 2NO$



VERDICT

Both changes are visible immediately from free elements $\rightarrow$ compound. This is redox.

MISCONCEPTION

Do not reject a reaction just because no explicit ions or electrons are written. Oxidation-number change is the criterion.

RR-C7

'Contains oxygen -> redox'

$\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$ contains oxygen but has no ON change.

'Combination -> redox'

$\text{C} + \text{O}_2$ is redox, but $\text{CaO} + \text{H}_2\text{O}$ is not.

'Gas forms -> redox'

Gas formation alone is not evidence; track oxidation numbers.

'Decomposition -> redox'

HgO decomposition is redox; MgCO_3 decomposition is not.

RETRY A

$3\text{O}_2 \rightarrow 2\text{O}_3$: redox or non-redox? State the oxidation-number evidence.

RETRY B

$\text{Zn} + 2\text{AgCN} \rightarrow 2\text{Ag} + \text{Zn(CN)}_2$: redox or non-redox? Track Zn and Ag.

Practice: scaffold fading

Prove the verdict with oxidation numbers.

RR-C8

Try with me

A. $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$

Ca +2 $\rightarrow$ +2; H +1 $\rightarrow$ +1; O -2 $\rightarrow$ -2.

No change $\rightarrow$ NON-REDOX.

B. $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$

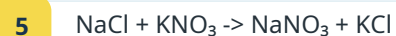
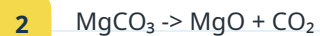
Hg +2 $\rightarrow$ 0 (down); O -2 $\rightarrow$ 0 (up).

UP + DOWN $\rightarrow$ REDOX.

PATTERN

The reaction label does not decide. The oxidation-number evidence decides.

Now you try



ANSWER STRIP

redox; non-redox; non-redox; redox; non-redox; redox.

Subtopic self-check

Recognition should now be evidence-based, not label-based.

AUDIT

Learner checklist

- 1 I can define a redox reaction by oxidation-number change.
- 2 I can prove non-redox when all tracked oxidation numbers stay unchanged.
- 3 I look for one UP and one DOWN in a complete redox reaction.
- 4 I know combination reactions can be redox or non-redox.
- 5 I know decomposition reactions can be redox or non-redox.
- 6 I do not use visible change, gas formation or oxygen presence as proof.
- 7 I can scan MCQ options using likely-changing atoms before full calculation.

SOURCE EXAMPLES COVERED

C/O₂, CaO/H₂O, HgO decomposition and MgCO₃ decomposition are all explicitly used as source-grounded recognition contrasts.

Completeness audit

SOURCE

PASS

PEDAGOGY

PASS

FAMILIAR CONTRAST

PASS

DECISION HELPER

PASS

MISCONCEPTION REPAIRS

PASS

SCAFFOLD FADING

PASS

CHEMISTRY TYPOGRAPHY

PASS

HEADER FAILURES

0

LAYOUT FAILURES

0

TRANSFER READY

PASS

NEXT

Next: classify redox reactions as combination, decomposition, displacement, disproportionation, comproportionation, intermolecular or intramolecular.

REDOX REACTIONS

SUBTOPIC 05 - TYPES OF REDOX REACTIONS

Classify by reaction topology only after proving the redox change

THE QUESTION

Once a reaction is known to be redox, can you classify it as combination, decomposition, displacement, disproportionation, comproportionation, intermolecular or intramolecular?

TWO LAYERS

Layer 1: prove redox from oxidation-number change.
Layer 2: classify the reaction pattern.

Redox status and reaction type are separate labels

First prove electron change; then inspect topology.

RT-C1

Why two labels?

The supplied source explicitly warns that combination and decomposition reactions are not automatically redox. So classification must happen in two stages.

STAGE 1 - REDOX?

Track oxidation numbers. If there is no UP and DOWN, stop: it is non-redox.

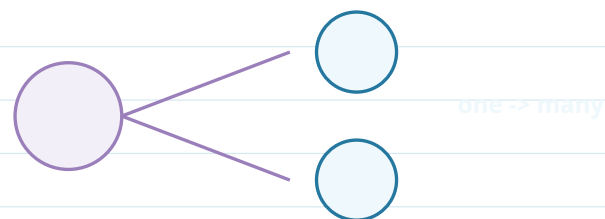
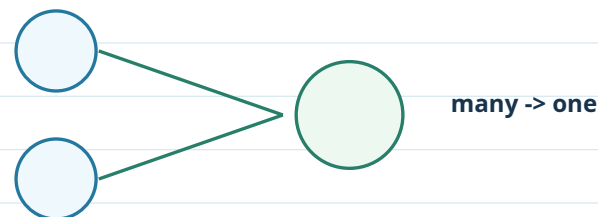
STAGE 2 - WHAT TYPE?

If redox is proved, inspect how reactants/products are arranged: combine, split, displace, split one oxidation state, or converge two states.

MISCONCEPTION

Wrong: 'decomposition' itself proves redox. Repair: reaction type is not redox evidence.

Topology helper



THINGS TO KNOW

Combination/decomposition describe number-pattern topology. Displacement describes one element replacing another. Disproportionation/comproportionation describe the same element's oxidation-state topology.

Combination and decomposition

The source teaches each with a redox and non-redox contrast.

RT-C2

Combination



TYPE + STATUS

Many reactants form one product and oxidation numbers change -> combination redox.

CLOSE CONTRAST

$\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$ is combination but non-redox.

Decomposition



TYPE + STATUS

One reactant forms several products and oxidation numbers change -> decomposition redox.

CLOSE CONTRAST

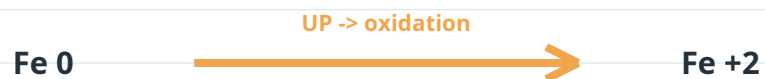
$\text{MgCO}_3 \rightarrow \text{MgO} + \text{CO}_2$ is decomposition but non-redox.

Displacement redox

An element replaces another element from a compound.

RT-C3

Source example: $\text{Fe} + \text{CuSO}_4$



TOPOLOGY

Fe enters the compound while Cu leaves as elemental Cu. One element has displaced another.

SOURCE RULE

The source treats displacement reactions in this section as redox reactions.

Second source example: $\text{Na} + \text{H}_2\text{O}$



CONCEPT HELPER

Look for a free element on one side and a different free element on the other. Then verify the oxidation-number swap.

MISCONCEPTION

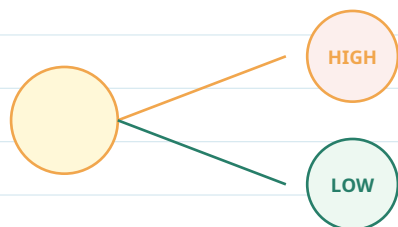
Do not call a simple ion exchange displacement redox unless an element is actually replaced.

Disproportionation

One oxidation state splits into a lower and a higher state.

RT-C4

The split pattern



CORE IDEA

The same element starts at one oxidation state and is both oxidised and reduced: MIDDLE $\rightarrow$ LOW + HIGH.

SOURCE EXAMPLE: Cl_2

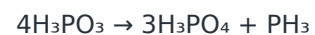
$\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaOCl} + \text{H}_2\text{O}$
Cl: $0 \rightarrow -1$ and $0 \rightarrow +1$.

VERDICT

Same starting Cl state moves both ways $\rightarrow$ disproportionation.

More source examples

H_3PO_3



P: $+3 \rightarrow +5$ and $+3 \rightarrow -3$.

Hg_2Cl_2



Hg(I) produces Hg(0) and Hg(II).

RECOGNITION TEST

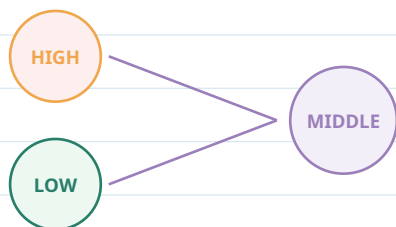
One reactant species + same element + one starting ON + two product ONs on opposite sides of the starting value.

Comproportionation

Two oxidation states converge to one intermediate state.

RT-C5

The convergence pattern



CORE IDEA

Two reactants contain the same element at different oxidation states and form one common intermediate state: $\text{LOW} + \text{HIGH} \rightarrow \text{MIDDLE}$.

SOURCE EXAMPLE: Pb / PbO₂

$\text{Pb} + \text{PbO}_2 + 2\text{H}_2\text{SO}_4 \rightarrow 2\text{PbSO}_4 + 2\text{H}_2\text{O}$
Pb: $0 \rightarrow +2$ and $+4 \rightarrow +2$.

VERDICT

Two Pb states converge to Pb(+2) $\rightarrow$ comproportionation.

Second source pattern

Ag²⁺ + Ag

$\text{Ag}^{2+} + \text{Ag} \rightarrow 2\text{Ag}^+$

Ag: $+2 \rightarrow +1$ and $0 \rightarrow +1$.

CLOSE CONTRAST

Disproportionation: one state splits.
Comproportionation: two states converge.

MISCONCEPTION CLINIC

Wrong: any reaction involving the same element twice is comproportionation.
Repair: two different starting ONs must converge to one common ON.

Split or converge?

Use oxidation-state topology, not equation length.

RT-C6

Side-by-side memory map

DISPROPORTIONATION

One starting oxidation state.

MIDDLE $\rightarrow$ LOW + HIGH

Same reactant species is both oxidised and reduced.

COMPROPORTIONATION

Two different starting oxidation states.

LOW + HIGH $\rightarrow$ MIDDLE

One starting state is oxidised; the other is reduced.

FAST QUESTION

How many starting oxidation states of the tracked element are present?

Try with me

A. $\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}$

Cu starts only at +1, then becomes +2 and 0.

One state splits $\rightarrow$ disproportionation.

B. $\text{Ag}^{2+} + \text{Ag} \rightarrow 2\text{Ag}^+$

Ag starts at +2 and 0, both become +1.

Two states converge $\rightarrow$ comproportionation.

RETRY CHECK

$\text{Pb}(0) + \text{Pb}(+4) \rightarrow \text{Pb}(+2)$: split or converge?

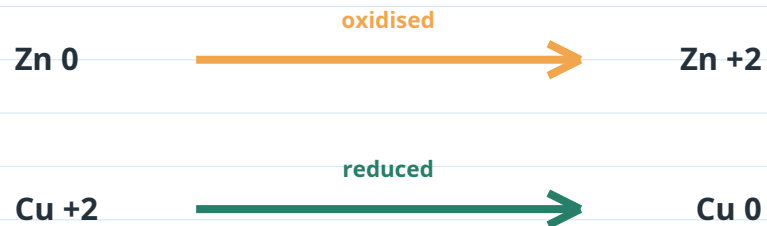
Intermolecular vs intramolecular redox

Ask where oxidation and reduction occur.

RT-C7

Intermolecular

The source defines intermolecular redox as oxidation of one reactant species and reduction of another reactant species.

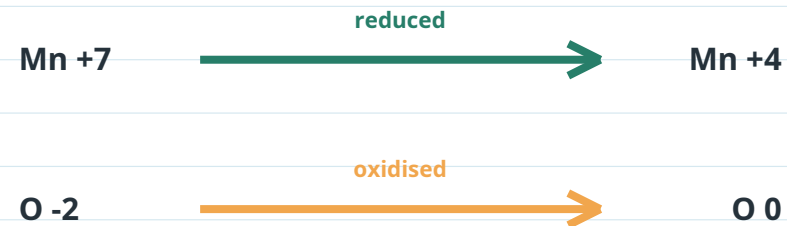


LOCATION TEST

Oxidation and reduction are carried by different reactant species -> intermolecular.

Intramolecular

The source defines intramolecular redox as oxidation and reduction occurring within the same reactant molecule/species.



LOCATION TEST

Both changing elements originate in the same reactant compound -> intramolecular.

Master classification workflow

Use pattern + oxidation-number evidence together.

RT-C8

Decision sequence

- 1 Prove redox: find an UP and a DOWN.
- 2 Many reactants $\rightarrow$ one product? Test combination.
- 3 One reactant $\rightarrow$ many products? Test decomposition.
- 4 Element replaces another element? Test displacement.
- 5 Same element: one ON $\rightarrow$ lower + higher? Disproportionation.
- 6 Same element: lower + higher $\rightarrow$ one ON? Comproportionation.
- 7 Finally ask location: different reactants = inter; same reactant =

IMPORTANT

A reaction may have more than one valid label, e.g. it can be both redox and displacement.

Quick classification set

- 1 $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$
- 2 $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$
- 3 $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$
- 4 $2\text{NO}_2 + 2\text{OH}^- \rightarrow \text{NO}_2^- + \text{NO}_3^- + \text{H}_2\text{O}$
- 5 $\text{Ag}^{2+} + \text{Ag} \rightarrow 2\text{Ag}^+$

ANSWER STRIP

combination; decomposition; displacement; disproportionation; comproportionation.

Misconception clinic + independent practice

Reaction type is topology supported by oxidation-state evidence.

RT-C9

'All combination reactions are redox'

False: $\text{CaO} + \text{H}_2\text{O}$ is combination but non-redox.

'Disproportionation means one molecule gives two products'

Not enough: the same element must move both lower and higher from one starting ON.

'All decomposition reactions are redox'

False: MgCO_3 decomposition has no ON change.

'Comproportionation is just the reverse equation shape'

Track the same element: two starting ONs must converge to one intermediate ON.

RETRY A

$\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}$: classify and justify using the Cu oxidation-state topology.

RETRY B

$\text{Pb} + \text{PbO}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{PbSO}_4 + \text{H}_2\text{O}$: split or converge?

Subtopic self-check

All source reaction-type families must be taught even if no direct PYQ exists.

AUDIT

Learner checklist

- 1 I separate redox status from reaction-type classification.
- 2 I can distinguish combination and decomposition using topology.
- 3 I can recognise displacement by element replacement and ON change.
- 4 I can identify disproportionation as one ON splitting lower + higher.
- 5 I can identify comproportionation as two ONs converging to one.
- 6 I can distinguish intermolecular from intramolecular by where changes occur.
- 7 I know one reaction can carry more than one valid label.

SOURCE COVERAGE

Combination, decomposition, displacement, disproportionation, comproportionation, intermolecular and intramolecular are all explicitly represented.

Completeness audit

SOURCE

PASS

PEDAGOGY

PASS

TOPOLOGY HELPERS

PASS

MISCONCEPTIONS

PASS

SCAFFOLD FADING

PASS

COMPROPORTIONATION

PASS

INTER / INTRA

PASS

TYPOGRAPHY

PASS

HEADER FAILURES

0

LAYOUT FAILURES

0

TRANSFER NOTE

Direct-primary JEE Main items cover combination/decomposition/displacement/disproportionation. No direct-primary item was found for comproportionation or inter/intra; Study Guide coverage closes those source obligations.

REDOX REACTIONS

SUBTOPIC 06 - INTEGRATED DECISION PIPELINE

Bring oxidation number, process, agents, redox recognition and reaction type into one method

THE FINAL EXECUTION SKILL

Given an unfamiliar equation, can you decide what to mark first, calculate only what matters, and stop at exactly the answer the question asks for?

NO NEW CHEMISTRY

This subtopic is a consolidation layer. It adds no topic beyond the supplied PDF; it integrates Subtopics 01-05 into one repeatable reasoning pipeline.

The canonical Redox reasoning pipeline

Use the same seven moves on unfamiliar equations.

IP-C1

P1 to P7

- P1 SELECT** atoms that matter
- P2 ASSIGN** oxidation numbers
- P3 TRACK** before after
- P4 CLASSIFY** oxidation / reduction
- P5 ATTACH** change to reactant species
- P6 NAME** oxidising / reducing agent

WHY THIS HELPS

The pipeline separates calculation from classification. Most agent mistakes happen when a learner skips directly from the equation to a label.

Where each earlier subtopic enters

P2 ASSIGN

Subtopic 01 - oxidation number

P3/P4 TRACK

Subtopic 02 - oxidation / reduction

P5/P6 ATTACH + NAME

Subtopic 03 - OA / RA

P7 REDOX CHECK

Subtopic 04 - redox / non-redox

TYPE if asked

Subtopic 05 - reaction classification

STOP RULE

Do not execute every stage if the question asks only for oxidation number or only for reaction type. Use the minimum necessary prefix of the pipeline.

Full worked example

One equation, all seven moves.

IP-C2



P1 SELECT

Track Zn and Cu. Sulfate remains unchanged for the redox decision.

P2-P3 ASSIGN + TRACK

Zn: $0 \rightarrow +2$
Cu: $+2 \rightarrow 0$

P4 CLASSIFY

Zn is oxidised; Cu^{2+} is reduced.

P7 CHECK

One UP and one DOWN -> internally consistent redox.

Attach, name and classify

P5 ATTACH

The oxidation belongs to reactant Zn. The reduction belongs to Cu^{2+} in CuSO_4 .

P6 NAME

Zn = reducing agent.
 Cu^{2+} / CuSO_4 = oxidising species / reagent label.

TYPE IF ASKED

Element Zn replaces Cu from a compound, so it is also a displacement redox reaction.

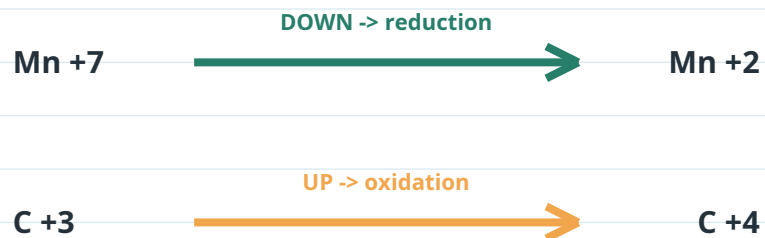
Harder equation: select only the changing atoms

Do not calculate oxidation numbers for every atom.

IP-C3

Permanganate + oxalic acid in acid

The source practice includes KMnO_4 with $\text{H}_2\text{C}_2\text{O}_4$ in acidic medium. The products show Mn ending at +2 and carbon ending in CO_2 .



MINIMUM PATH

Select Mn and C only $\rightarrow$ track $+7 \rightarrow +2$ and $+3 \rightarrow +4$ $\rightarrow$ classify $\rightarrow$ attach to KMnO_4 and $\text{H}_2\text{C}_2\text{O}_4$.

RESULT

$\text{KMnO}_4 = \text{OA}$; $\text{H}_2\text{C}_2\text{O}_4 = \text{RA}$.

What not to do

OVER-CALCULATION

Do not assign oxidation numbers to K, H and every O atom when the products already reveal the changing Mn and C atoms.

MEMORISED-REAGENT SHORTCUT

Knowing KMnO_4 is often an oxidant may help, but the equation should still be able to prove the role from $\text{Mn} +7 \rightarrow +2$.

CONCEPT HELPER

On a long equation, circle only species whose tracked element appears at a different oxidation state in the products.

Stop at the answer the question asks for

Not every question needs the full pipeline.

IP-C4

Question type -> stopping point

Find oxidation number

Stop after P2

Oxidised or reduced?

Stop after P4

Identify OA / RA

Continue through P6

Redox or non-redox?

Use P2-P4 + P7

EXAM SPEED RULE

The fastest correct solution is the shortest justified path, not the longest calculation.

Three mini examples

ONLY ON

KMnO_4 : K +1, O -2 -> Mn +7. Stop.

ONLY PROCESS

$\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$: +2 -> +3 -> oxidation. Stop.

ONLY TYPE

$\text{Ag}^{2+} + \text{Ag} \rightarrow 2\text{Ag}^+$: +2 and 0 converge to +1 -> comproportionation.

The six traps that still cause mixed-question errors

Repair the model, not just the answer.

IP-C5

P2: O/H exception

O is not always -2; inspect peroxide/superoxide/O-F.

P4: process reversal

ON UP = oxidation; ON DOWN = reduction.

P2: average vs actual

A fractional average may hide non-equivalent atoms.

P5/P6: agent inversion

Oxidised species = RA; reduced species = OA.

P7: false-redox trigger

New product / gas / oxygen does not prove redox.

TYPE: split vs converge

One ON splits = disproportionation; two ONs converge = comproportionation.

Source-question readiness check

Use the supplied PDF questions as the first validation corpus.

IP-C6

Directly supported source question families

- 1 $\text{Zn} + \text{HCl}$: identify the reducing agent.
- 2 $\text{KMnO}_4 / \text{H}_2\text{C}_2\text{O}_4$: identify the reductant.
- 3 Identify oxidation from electron loss / de-electronation.
- 4 $\text{Hg}^{2+} / \text{Sn}^{2+}$ reaction: identify which species is oxidised and reduced.
- 5 $\text{Ag}^{2+} + \text{Ag} \rightarrow 2\text{Ag}^+$: identify the reaction type.
- 6 $\text{Ca} + \text{Cl}_2 \rightarrow \text{CaCl}_2$: identify oxidant and reductant.

STATUS

These are supported by the P1-P7 pipeline plus the reaction-type extension.

One source item needs an external prerequisite

STRONGEST REDUCING METAL

The supplied practice asks for the strongest reducing agent among metals such as Ba, Mg, Al and K. Choosing among those metals requires metal reactivity / periodic knowledge not developed by the Redox rules in the supplied PDF.

HOW WE HANDLE IT

Do not pretend the P1-P7 pipeline alone can rank those metals. Mark the question as external-prerequisite for this learning package rather than silently teaching a new periodic-trend unit.

GROUNDING RULE

The source determines scope. A source question can still expose a prerequisite gap that the source theory itself did not teach.

A practical exam workflow

Use progressive depth instead of doing everything every time.

IP-C7

10-second scan

SCAN

1. Circle free elements / obvious changing ions.
2. Mark only likely changing oxidation numbers.
3. Look for UP / DOWN.

DECIDE

If the question asks process, stop at oxidation/reduction. If it asks agent, attach to reactant species. If it asks type, inspect topology after redox is proved.

CHECK

One UP should be paired with one DOWN in a complete redox equation.

When the scan is not enough

SLOW DOWN FOR

peroxide/superoxide/O-F oxygen; mixed/average oxidation states; ions with non-zero total charge; same-element split/converge questions.

DRAW A HELPER

Use a before→after lane, SELF/OTHER frame, charge-balance equation, or split/converge topology. A quick representation is often faster than mental juggling.

TARGET

Accuracy first, then speed. The same pipeline should survive unfamiliar formulas.

Cumulative check

Use the minimum justified path.

IP-C8

Mixed problems

- 1 Find Cr in $\text{Cr}_2\text{O}_7^{2-}$.
- 2 $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$: oxidation or reduction?
- 3 $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$: identify OA and RA.
- 4 $3\text{O}_2 \rightarrow 2\text{O}_3$: redox or non-redox?
- 5 $\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}$: classify the redox type.
- 6 $\text{Ag}^{2+} + \text{Ag} \rightarrow 2\text{Ag}^+$: classify the redox type.

ANSWER STRIP

Cr +6; oxidation; H^+/HCl OA and Zn RA; non-redox; disproportionation; comproportionation.

Explain, don't only label

- 1 Why is the oxidising agent itself reduced?
- 2 Why is $\text{O}_2 \rightarrow \text{O}_3$ not redox?
- 3 Why is +8/3 in Fe_3O_4 an average rather than every Fe site?
- 4 Why is $\text{Cl}_2 + \text{NaOH}$ a disproportionation reaction?
- 5 Why can a reaction be both redox and displacement?

SUCCESS CRITERION

A correct answer should include the evidence: oxidation numbers, direction of change, species attachment, or topology.

Subtopic self-check

This is the integration gate before chapter-level closeout.

AUDIT

Learner checklist

- 1 I can use P1-P7 without guessing agent labels.
- 2 I know where to stop for each question type.
- 3 I can select changing atoms instead of calculating every atom.
- 4 I can switch to a helper for exceptions, averages or split/converge cases.
- 5 I can explain my answer using oxidation-number evidence.
- 6 I can identify when a question needs knowledge outside the supplied Redox scope.

NO NEW SOURCE OBLIGATION

Subtopic 06 integrates existing obligations only. It does not add balancing, titration, electrochemistry, or reaction-specific inorganic chemistry.

Integration audit

P1 SELECT

PASS

P2 ASSIGN

PASS

P3 TRACK

PASS

P4 CLASSIFY

PASS

P5 ATTACH

PASS

P6 NAME

PASS

P7 CHECK

PASS

TYPE EXTENSION

PASS

SOURCE-Q SUPPORT

PASS

SCOPE LEAKS

0

NEXT

After the cumulative ExamSIDE challenge, the remaining task is a chapter-level audit / merge proving every source obligation and every eligible canonical PYQ has a unique home.

Chapter closeout

Source content complete; external corpus complete after RX-ST05 backfill.

CLOSEOUT

SOURCE CONTENT

20 / 20 source obligations taught

No new chemistry introduced from ExamSIDE.

EXAMSIDE CORPUS

74 candidates audited

24 eligible • 8 partial • 42 out-of-scope

Eligible primary placements = 24 / 24.

SOURCE INTERNAL QUESTIONS

All direct Redox-process questions are supported. One printed 'strongest reducing metal' item remains a documented external-prerequisite exception rather than triggering a new periodic/reactivity unit.

PRIMARY PLACEMENT

RX-ST01 7/7

RX-ST02 2/2

RX-ST03 4/4

RX-ST04 3/3

RX-ST05 8/8

Duplicate primary placements = 0.

PEDAGOGY / QA

Each core subtopic includes helpers, misconceptions, worked/guided/faded practice, typography QA and layout QA.

RX-ST06

Cumulative challenge reuses canonical PYQs only;
NEW_PRIMARY_REQUIRED = 0.